



The role of IT leadership in realizing the return on investment of AI coding assistants.

by

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Abstract

The rapid adoption of generative artificial intelligence (AI) coding assistants is changing how software engineering teams develop, test, document, and deliver software. Despite previous research showing that AI coding assistants can be effective at improving developer productivity, it does not follow that such improvements will result in better ROI. The benefits of integrating AI in a company also depend on how well it manages, promotes and integrates the technology in the routine software development process. This study investigates how organizational culture and strategic leadership influence the realization of ROI from AI coding assistants in software engineering teams. The study is guided by Technology Value Realization Theory as well as the Socio-Technical Systems perspective, which provide a basis for understanding how technological benefits depend on both organizational and human factors.

This research employs the quantitative research design to explore the linkages among various factors like organizational culture, strategic leadership, AI coding assistant assimilation quality, and the return on investments. The study focuses on software engineering teams of medium-scale and large organizations in South Africa that have implemented the AI code assistant technology. It has been planned to collect the data through a well-structured survey with the help of the Likert scale. The return on investment in case of AI coding assistant has been examined on three levels, which are the impact at the level of task, the impacts regarding integration and coordination at the level of workflow, and the effects on the financial and futuristic aspect of the company. AI coding assistant assimilation quality is considered to be the mediating variable along with organizational readiness, the maturity and experience of the team, and length of the adoption as the control variables.

The study aims to fill the gap in existing body of knowledge by examining opportunities to realize profits from the usage of AI programming assistants and the impact of organization-

related issues on the overall productivity of programmers. The research will use statistical approaches, e.g., Structural Equation Modeling, to establish the reliability and validity of the measurement model as well as test the relationships between variables in the study. The research is expected to determine if organizational culture and strategic leadership directly impact AI-related ROI and contribute to the proper assimilation of AI coding assistants into software engineering processes.

The study contributes to AI value realization research by linking technology performance to organizational culture and leadership. The study results are expected to provide practical tips for software engineering managers and companies aiming to deploy AI coding assistants in the way that would guarantee positive productivity and business results.

Keywords: Artificial intelligence, AI coding assistants, organizational culture, strategic leadership, technology value realization, assimilation quality, return on investment, software engineering.

Declaration

I, Celumusa H Zulu, certify that the dissertation submitted by me for the Master of Commerce in Information Technology Management degree at the University of Johannesburg is my independent work and has not been submitted by me for a degree at another university.

CELUMUSA H ZULU

(No signature required)

Dedication

To my Mother, who always believe in me, my kids and to all my Siblings who always believe in me.

To Evelyn Manosa thank you very much for everything you did to make this study possible.

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First and foremost, infinite gratitude and appreciation to my Creator, Who has given me the will to start studying again and the endurance to see this through in a time when my world was turned on its head. My perseverance, accomplishments and very being are only by Your will.

Also, sincerest gratitude to my Kids, who had to endure my absence whilst completing this study. You are the very foundation my success is built on.

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Chapter 1

Introduction

1.1 Introduction

This chapter introduces the study and also provides the background for examining The role of IT leadership in realizing the return on investment of AI coding assistants. It outlines the research problem being addressed, presents the research objectives and research questions of this study. Furthermore the chapter also discusses the significance of the topic and also states the scope and delimitation. Lastly, this chapter will provide the theoretical framework used to guide the study and close off with the dissertation structure.

1.2 Background

Artificial intelligence (AI) is one of the structural factors that has a significant impact on productivity and competitive advantages (Gillespie et al., [2025](#)). As of 2023, generative AI systems have developed from being an experiment to reaching the phase of implementation within organizational processes (Gillespie et al., [2025](#)). Recent reports from around the world show that organizations allocate more money to AI since they expect clear monetary outcomes, cost savings, and improved performance of organizational processes (Gillespie et al., [2025](#)). Research on information systems and management has transcended from the stage of adoption to the understanding of added value meanings with the emphasis on differentiation of organizational value created from investments in AI systems as a function of organizational governance and management performance, not technology excellence (Hossain et al., [2025](#)).

In software engineering, the use of generative AI tools are among the most noticeable forms of AI-enabled augmentation (Peng et al., 2023). Providing developers with functionalities such as code suggestions and entire code functions in real time based on context, documentation, and debugging (Peng et al., 2023). Practical controlled experiments provide conclusive evidence that AI-assisted developers complete software development tasks faster and report increased productivity of about 55.8% compared with control groups (Peng et al., 2023). Collaboration between humans and AI systems is a paradigm shift from the traditional concept of automation, in which AI enhances developers' capabilities rather than replacing them with machines (Peng et al., 2023). Initial performance improvements demonstrated in controlled promising experiments may not necessarily guarantee long-term organizational performance benefits. Empirical studies report considerable variation in the productivity gains from AI-assisted software development tasks across software development teams and organizations.

As it has been long noted, organization culture and management style have been generally believed to affect the implementation success of new technologies. Within the digital field, values and practices will drive the culture of experimentation, learning, and data use in decision making as well as a positive attitude to tech adoption. New findings reveal that digital culture explains the linkage between human AI skills and innovation outcomes, and that culture readiness mediates the performance impact of AI Investments (An et al., 2024). Similarly, this applied also to employees' engagement in AI, ethical AI management or leadership impacts toward strategic alignment. Even though the significance of digital culture and leadership is being gradually recognized by most researchers, current research findings on AI Adoption is more fragmented (An et al., 2024). A significant number of recent works investigated AI adoption under the macro-organizational view or focused on the impacts for personal performance only instead of integrating digital culture, management and ROI in the software industry (Peng et al., 2023).

Generative AI (GenAI) has been increasingly adopted by African companies to address resource constraints and operational inefficiencies. Similarly, there is evidence that developing nations have tried extensively with AI and have positive views about AI's capabilities to boost productivity at work (Gillespie et al., 2025). Yet, literature to date highlights difficulties around digital skills pipelines, governance maturity and leadership capability among others in adopting GenAI technologies (Hughes et al., 2026). In this context, organizational culture and leadership remain particularly critical. If these factors are not supported by a clear strategic vision, trust

and a learning orientation, AI adoption is likely to be less about value creation and more about segmentation or symbolism. The digital transformation landscape in Africa is a valid environment for exploring the impact of socio-organizational factors on the achievement of returns from AI technologies.

With regards to software engineering in South Africa, the environment provides a relevant framework in grasping how the various organizations adopt and utilize advanced AI tools. Several studies confirm the fact that AI has emerged as one of the major components that define the transformation of operations on companies and the way they develop their culture due to the increase in number of firms using AI technologies in their business processes (Murire, 2024). There are opportunities and challenges that needs to be addressed in order to leverage organizational performance through AI adoption and alignment with organizational culture and structure, as indicated by research reviews on AI tool adoption in organizations (Romeo & Lacko, 2025). Though the adoption of AI technology affects many aspects of the organization's performance positively, like productivity and output, the effect of AI technology on aspects like the organization's culture and employees morale and engagement is complicated and depends on many factors, as indicated by the study on the impact of AI tools in workplace settings (Kassa & Worku, 2025). Studies on the effect of generative AI tools on software engineering have indicated that AI technology positively affects organizations by enhancing productivity in coding and other software engineering activities through employees adoption of AI tools (Alenezi & Akour, 2025).

Leadership and organizational culture play an important role in the interpretation, adoption and usage of AI technologies by work teams. Research on AI adoption has found that the adoption of AI related technology is also dependent on leadership styles that facilitate embedding technology initiatives into organizational culture (Romeo & Lacko, 2025). Research has also found that the adoption of AI-related technology is dependent on the adoption of the organizational culture that is conducive to the adoption of technology (Kassa & Worku, 2025). Further more research on organizational culture has also shown that cultural elements, particularly the norms that define innovation, collaboration and adaptability, have an impact on the use of AI by staff (Murire, 2024). In the case of software engineering teams that use AI-based coding assistants for various functions, both leadership as well as culture determine the effectiveness of AI in terms of productivity or quality and financial returns.

1.3 Problem statement

Generative AI was primarily used for research and information retrieval purposes in its initial years; however, the use of generative AI has shifted towards the process of knowledge work itself. Experimental results show that generative AI enhances the efficiency and effectiveness of cognitive work (Noy & Zhang, 2023). Furthermore, studies conducted in a work environment setting have also shown improvements in productivity as well as employees performance due to the use of generative AI. It seems from these results that the benefit of generative AI depends upon its implementation rather than the technology itself (Brynjolfsson et al., 2025). More experimental results regarding software development can be found in favour of this theory. Large-scale researches conducted on GitHub Copilot have been showing improvements in the completion time as well as in the efficiency of the development process of developers (Cui et al., 2026). Furthermore, controlled experiments using AI programming assistants have also showed improvements in the speed of completing software development tasks, this provides empirical support for the ROI of generative AI in coding assistance (Peng et al., 2023). All these results collectively show that AI coding assistants are capable of increasing productivity. It is therefore necessary to conduct more research in order to investigate the effects of organizational factors on this phenomenon.

While all these improvements are measurable, research evidence in existing literature has proved that the performance impact varies based on the situation. In software engineering, research has been able to demonstrate that the code generated using GitHub Copilot increases productivity but at the same time creates a level of inconsistency in accuracy, maintainability, and effort required for review purposes; and this clearly demonstrates the need to depend on the management and integration of the system (Moradi Dakhel et al., 2023). Research on the governance of artificial intelligence in organizations has suggested that to achieve the sustainability of value creation, there should be processes and systems in place to ensure accountability and supervision of the leadership to convert experiments into performance improvement (Mäntymäki et al., 2022). Further more studies on organizational innovation with generative AI has demonstrated that the organization will be able to achieve various performance impacts depending on whether the AI is used for automation or augmentation purposes; and this demonstrates the importance of strategic leadership and culture (Holmström & Carroll, 2025). Overall, this existing research evidence shows the role of organizational culture and strategic leadership in determining whether

the improvements in productivity due to AI coding assistants can be sustained and converted to measurable return on investment.

The importance of such mechanisms can also be seen within the context of Africa due to the fact that organizations face limitations in terms of skills development, governance, and digitization. The empirical study on the readiness of AI within Africa demonstrated the level of readiness and the organization's ability to implement AI which demonstrates the importance of organizations in the transition from AI pilot to its performance (Isagah & Ben Dhaou, 2024). In the context of South Africa, empirical studies on the financial sector have demonstrated the role of organizational factors, including leadership support and readiness in the adoption of new and emerging technologies which demonstrate the importance of such factors in achieving ROI from using AI coding assistants in software development teams (Matsepe & Van der Lingen, 2022). The empirical research conducted in South African environment highlights the significance of sociotechnical systems to AI adoption and change management as mechanisms of achieving performance outcomes related to AI adoption (Smit et al., 2024). Also the research reveals the significance of managerial perceptions towards AI adoption in achieving performance outcomes within the field of human resources management as the field of AI application.

Collectively, the global evidence shows that AI coding assistants can improve productivity, while organisational research suggests that sustained ROI is influenced by strategic alignment, effective governance, and organisational culture. The African and South African evidence base also indicates that the context is significant in relation to the successful adoption of technology. It is because of this that the evidence base offers a rationale for conducting an in-depth exploration in relation to the role of culture and leadership in achieving ROI with AI coding assistants.

1.4 Research aim and objectives

1.4.1 Main Research Objective

To develop an integrated framework linking financial metrics, organizational culture, and leadership to assess and enhance the return on investment of AI coding assistants in South African software engineering teams.

1.4.2 Sub Research Objectives

1. To identify and operationalize the financial costs and benefits that define return on investment for AI coding assistants.
2. To examine how organizational culture attributes influence productivity and quality outcomes from AI coding assistants.
3. To assess which leadership practices most strongly affect return on investment outcomes.
4. To apply and validate an integrated return on investment framework within large South African organizations.
5. To formulate practical recommendations that enable managers to maximize return on investment through cultural and leadership interventions.

1.5 Research questions

1.5.1 Main Research Question

How do organizational culture and leadership shape the return on investment of AI coding assistants in South African software engineering teams?

1.5.2 Sub Research Questions

- RQ1. What financial costs and benefits define return on investment for AI coding assistants in software engineering teams?
- RQ2. How do organizational culture attributes influence productivity and quality outcomes from AI coding assistants?
- RQ3. Which of the leadership behaviours strongly affect return on investment outcomes from AI coding assistants?
- RQ4. How can financial and organizational factors be integrated into a practical return on investment framework for large South African organizations?
- RQ5. What practical guidance can be offered to managers to maximize returns from AI coding assistance investments through cultural and leadership interventions?

1.6 Significance/rationale

Generative AI has a potential to enhance productivity in software engineering and efficiency in knowledge work (Brynjolfsson et al., 2025; Peng et al., 2023). AI coding assistants speed up the software development process and make the completion of the task more probable. Nevertheless, even though the productivity has increased, the connection between the use of AI coding assistants and the ROI obtained by software engineering teams remains unclear.

Leadership and organizational culture have been identified as factors that affect the successful adoption of the associated benefits of using AI technology. It affects innovation through guiding organizational resources towards innovative practices (Hossain et al., 2025). A culture of learning, cooperation, and innovation is essential for organizational digital capability (Abdul Wahab & Radmehr, 2024; Li et al., 2025; Poisat et al., 2024). A system perspective, adaptability, is essential for organizational assimilation of digital technologies. Performance of emerging markets is dependent on their level of readiness and leadership context (Gillespie et al., 2025; Nurlia et al., 2023).

However, a link still seems to be missing since there is no literature link between organizational ROI and the productivity of AI coding assistants. Likewise, there is no literature link between organizational ROI and AI value. Finally, the literature on organizational culture and leadership does not have any literature link between organizational ROI and AI coding assistants' value for software engineering teams. This study fills in this gap since it will explore the influence of organizational culture and leadership on ROI of AI coding assistants for software engineering teams.

1.7 Scope and delimitation

1.7.1 Scope

This study is guided by a quantitative research approach that seeks to explore the impact of organizational culture and leadership on the achievement of return on investment (ROI) for AI coding assistants in software engineering teams. The study is limited to the usage of generative AI tools that facilitate the inclusion of code, inline suggestions, debugging, and documentation, as they are part of the software development process. Literature has already proven that the usage of such tools increases productivity as well as the efficiency of task

completion (Brynjolfsson et al., 2025; Noy & Zhang, 2023; Peng et al., 2023).

The target population is software engineering teams working for medium-to-large-scale organizations in South Africa who are using AI coding assistants tools. The reasonable time frame for evaluating the outcomes varies for each company since the adoption of AI coding assistants is dependent on firm-level characteristics (Babina et al., 2024). Moreover, the digital readiness of the African continent is recognized as an environmental factor (Gillespie et al., 2025).

The scope of independent variables is limited to organizational culture and leadership in this study, as they might impact the achievement of return on investment for AI coding assistants. The organizational culture is analyzed by taking the learning orientation, collaboration, and innovation support as factors (Abdul Wahab & Radmehr, 2024; Fosso Wamba et al., 2024). Leadership is analyzed by taking the strategic direction, governance, and digital support as factors (An et al., 2024; Hossain et al., 2025; Murire, 2024). The return on investment is analyzed by taking the quantified performance indicators as factors.

1.7.2 Delimitation

Because the technical efficiency of AI tools has already been studied in experiments (Peng et al., 2023), it is important to highlight that this research does not focus on the accuracy benchmarking, technical algorithmic evaluation of the efficiency of the AI coding assistants, architecture of the AI model, or other technical evaluations of the efficiency. While in this research, there was a choice to focus on the socio-organizational factors of value realization in the context of AI. The research only includes software engineering teams, not other professional groups using generative AI, such as marketing, finance, or legal teams. This is because the software engineering workflow is significantly different from other professions.

The significance of leadership and organizational culture is considered in the study as the main independent elements of the research. However, some variables which also could have an impact on ROI, including national regulations, macro environment, and pricing models have not been analyzed properly. While it is evident that those variables can considerably influence ROI, the objective of the current research is to focus on only internal variables.

1.7.3 Brief Limitations

There are several constraints that are inherent in this study. To begin with, the calculation of ROI will rely somewhat on self-reported perceptions of productivity and quality. Secondly, the study is limited in terms of violations of generalizability by virtue of its focus on South African organizations. Lastly, the cross-sectional or short-term longitudinal aspect of the design will be without power in terms of the long-term impact on financial results because the effect of the culture and leadership on the productivity might take a long time to manifest itself(An et al., 2024).

1.8 Theoretical and conceptual framework

This study is based on the following premise: while the productivity gains derived from the use of AI coding assistants can be significant for developers, the value that can be derived from the productivity gains for the organization itself remains an open question. This view is consistent with the value realization framework, which argues that business value depends on how the organization manages, supports, and integrates the technology into its day-to-day business processes. It can also connect with the sociotechnical systems perspective noting that, value derived from the utilization of AI coding assistants depends on the people, processes, leaders, and organizations involved.

There is a significant body of research that suggests that generative AI has a positive impact on individual performance. Research conducted in the field has shown that generative AI has a positive effect on the productivity and quality of performance (Brynjolfsson et al., 2025). Experimental research has also shown that generative AI has a positive effect on the efficiency and quality of performance (Noy & Zhang, 2023). Research has shown that programmers working with AI assistants perform their tasks more efficiently and with better quality than those programmers who do not use AI assistants (Peng et al., 2023). However, research conducted at the firm level has shown that the ROI generated by AI is firm-specific, depending upon how the firm invests in AI (Babina et al., 2024).

In this context, leadership and organizational culture are identified as key factors that explain the disparities identified. Leadership is responsible for guiding and directing the firm, as well as the way AI coding assistants are used within the firm. The empirical evidence suggests that

leadership is a key factor that affects digital or AI performance outcomes (Hossain et al., 2025; Murire, 2024). On the other hand, organizational culture is responsible for influencing the behavioural aspects of individuals within the firm. Learning culture is a culture that promotes the use of AI tools within the firm through collaboration and innovation. The empirical evidence suggests that a digital organizational culture strengthens the relationship between AI performance outcomes and organizational performance (Fosso Wamba et al., 2024). Furthermore, a digital organizational culture increases innovation outcomes by human-AI capabilities (An et al., 2024).

The conceptual model in question defines ROI from AI coding assistants as the key variable of particular interest are the roles of IT leadership and organizational culture in relation to ROI as the explaining variables in the model. In addition, the importance of assimilation quality is explained in the context of the mediating role of leadership and organizational culture in their effect on ROI. Assimilation quality is understood as the stage in which AI coding assistants are integrated into the process of software engineering and start being used in existing workflows. Furthermore, organizational readiness is included in the model as a control variable, the potential for successful implementation of AI coding assistants would depend on the level of organizational readiness to adopt and work with digital technologies in general. Even though there is available literature reporting on higher performance on the level of individuals and tasks, little is known about the relationship between the achieved productivity level and the ROI from AI coding assistants on the level of organizations (Gillespie et al., 2025; Nurlia et al., 2023).

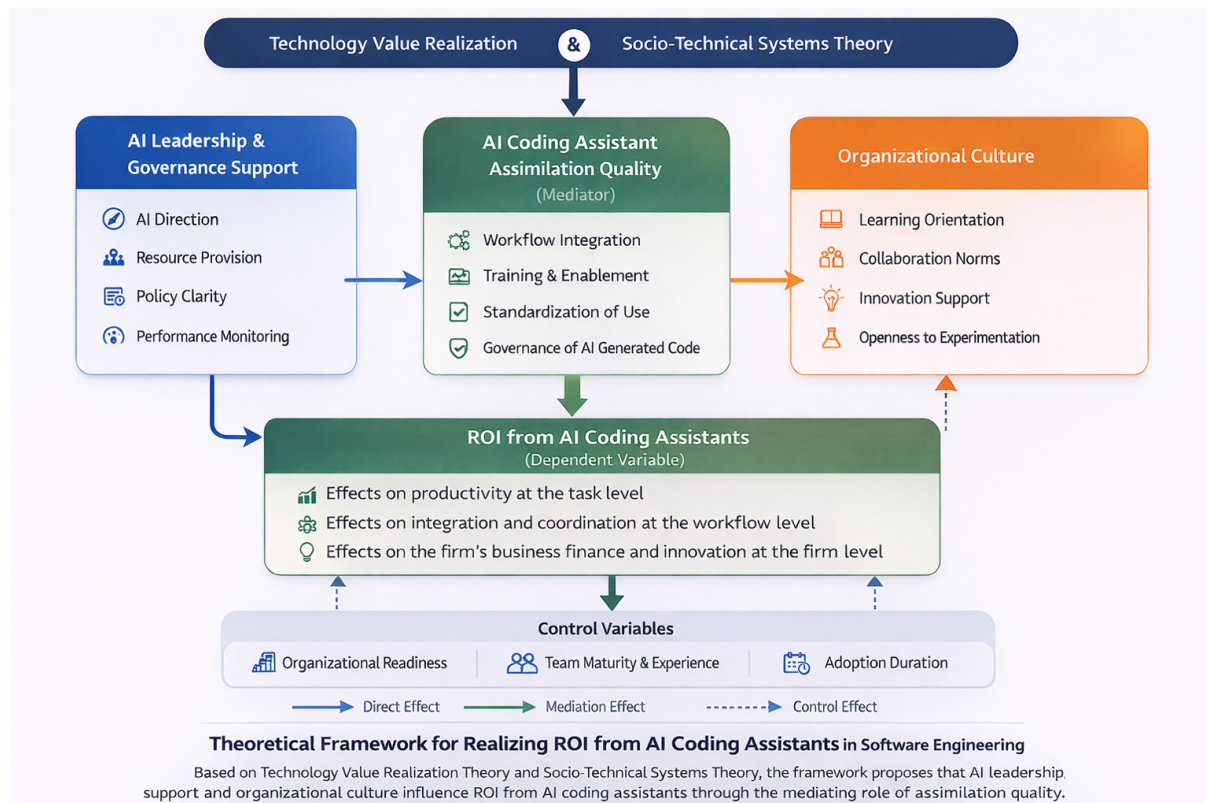


Figure 1.1: theoretical framework.

1.9 Dissertation structure

This dissertation has about 6 chapters outlined below and in Figure 1.2:

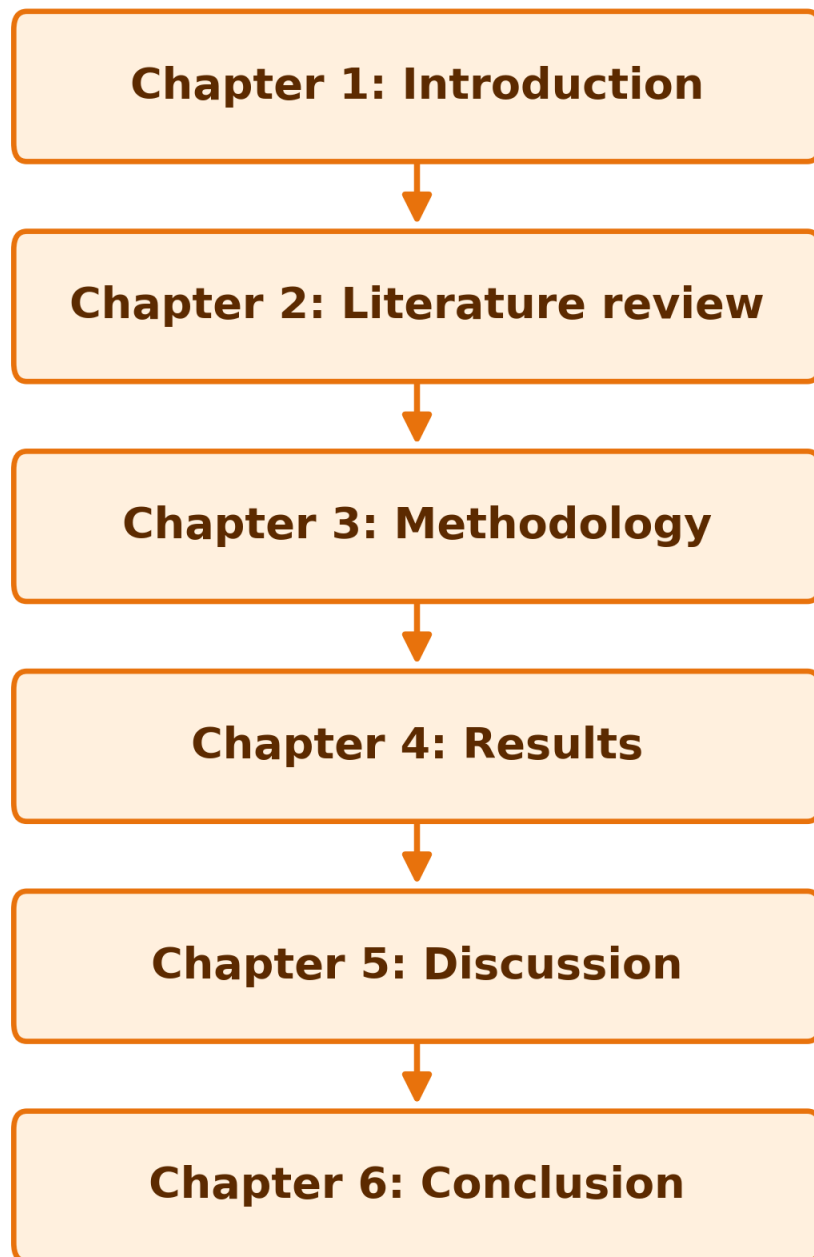


Figure 1.2: dissertation structure.

Chapter 1: Introduction

Chapter 1 introduces the study by providing the background for examining The role of IT leadership in realizing the return on investment of AI coding assistants. It outlines the research problem being addressed, presents the studies objectives and research questions of this study. Furthermore the chapter also discusses the significance of the topic and also states the scope and delimitation. Lastly, this chapter provides the theoretical framework used to guide the study and closed off with the dissertation structure.

Chapter 2: Literature review**Chapter 3: Methodology****Chapter 4: Results****Chapter 5: Discussion****Chapter 6: Conclusion****1.10 Chapter summary**

Chapter 2

Literature Review

2.1 Introduction

Generative AI has become one of the most used technologies in different fields to perform various tasks where high levels of knowledge and information are required. This is different from traditional AI in the sense that this technology emphasizes cognitive complexity within the output created and responding to the need for community-specific interpretation, refinement, and integration. In software development, artificial intelligence is used in the form of coding-monitoring assistant tools that are included in the software-development environment to facilitate code generation, fixing bugs in the software, and creating documentation.

Empirical studies on the impact of generative AI on human productivity and output quality revealed that, under conditions of structured task design, a significant increase in human output is achieved. In a large-scale experimental design, the availability of generative AI tools resulted in a reduction on task completion time and an increase in output quality. The results achieved were statistically significant for the total sample, while the results were more pronounced for low-performing individuals (Noy & Zhang, 2023). The results support the view that generative AI tools can produce productivity gains at the individual task level.

However, the productivity gains at the task level do not necessarily translate into organizational return on investment. Research on AI investment at the firm level indicates that the performance effects are heterogeneous and conditional. AI investment is linked with growth through innovation and product development channels; however, the level and stability of the returns

depend on the capabilities of the organization (Babina et al., 2024). The market reaction to AI investment announcements is mixed, indicating the level of awareness among investors about the risk and capabilities (Lui et al., 2021).

These studies all circle back to the same basic idea: AI capability does not produce value; rather, value is produced when the technological tools are embedded in the organizational systems. More recent studies have suggested another component: a digital organizational culture can influence the innovation outcome of AI capability (An et al., 2024). Leadership can also play a part in influencing the development of trust and embedding AI usage (Schneider et al., 2023).

Despite the recent increase in research on the performance effects of AI, little research has appeared that combines the effects of organizational culture and leadership into a structured ROI model for AI coding assistants in software development teams. This chapter synthesizes global empirical research on AI productivity and firm performance, critically evaluates the impact of organizational conditions, and focuses the research question more specifically on the South African market, where empirical ROI research has yet to be conducted.

2.2 Generative AI and Productivity: From Task Performance to Organizational Outcomes

The best evidence for how generative AI changes work comes from tightly controlled experiments. In the case of Noy and Zhang (2023), the evidence provides strong causal proof of how gen AI increases the rate of task completion and enhances the quality of professional writing. From this, we can see precisely how the presence of gen AI changes the gap in performance. The benefits of using gen AI are not equally distributed, with the less skilled benefiting the most, this suggests that generative AI has the potential to narrow productivity differences between teams..

This uneven benefit to different individuals can, therefore, have implications for software development teams. If the benefits to gen AI for coding tasks are greater for the less skilled, the composition of the team might change. However, the benefits to the team must not only be seen at the task level. They must be sustained, integrated, and monetized to increase the ROI.

This difference is supported by firm-level evidence. According to Babina et al. (2024), adoption of AI has a positive relationship with both firm growth and innovation, and the relationship

is more pronounced for firms equipped with complementary capabilities. The association is believed to be realized mainly via product innovation, as opposed to labor replacement effects, thus suggesting that AI assists in value creation through product innovation rather than reducing costs.

At the same time, there is a clear indication regarding mixed responses of financial markets concerning announcements stating the investment in artificial intelligence. Lui et al. (2021) report that there have been firms exhibiting positive abnormal profits on the date AI investment announcements are made and others showing neutral and negative returns. This appears to reflect investors ability to distinguish between firms with the ability to successfully implement AI and those with a risk of implementation failure.

In the context of software engineering, empirical evidence points to the fact that there is a great deal of usage of AI coding assistants. However, this usage is done in an inconsistent manner. In a comprehensive empirical study done to examine the experiences of software developers, it was found that there was usage of AI coding assistants for various activities in the software development process. However, issues of trust, boundaries, and governance ambiguity have been found to hinder the effective impact of this usage (Sergeyuk et al., 2025). This points to the fact that the frequency of usage alone may not guarantee the achievement of benefits.

According to the literature, the return on investment (ROI) model of AI coding assistants comprises a three-tier structure:

2.2.1 Effects on productivity at the task level.

Task-level productivity effects relate to the observed improvements that accrue when individual programmers utilize AI-based coding assistants for the accomplishment of specific programming tasks. At the task level, AI-based coding assistants are used for activities such as code writing, documentation, debugging, and suggestion of solutions to programming problems. Such activities have the potential to fast-track the accomplishment of specific tasks through the provision of real-time support during the entire process. According to experimental research, generative AI-based tools have the capability to aid the execution of knowledge-based tasks by a large factor without compromising the quality of the output (Noy & Zhang, 2023). This implies that, at the organizational level, the ability of the development team to quicken the accomplishment of specific coding tasks has the capability to accelerate the accomplishment of more complex

development tasks.

2.2.2 Effects on integration and coordination at the workflow level.

Workflow-level integration and coordination effects refer to the impact of AI coding assistants in the collaboration and coordination of teams in the development of software applications. Software development involves the implementation of workflows that include code reviews, testing procedures, version control, and collaboration between different developers. The incorporation of AI coding assistants in the development process can have positive effects on the coordination and collaboration of teams in the development process. This means that the teams can be able to achieve the required objectives in the most efficient manner possible. The benefits of using AI coding assistants have been examined in the experiences of developers using the system, and the findings suggest that the benefits of the system depend on the extent of its integration in the development process (Sergeyuk et al., 2025).

2.2.3 Effects on the firm's business finance and innovation at the firm level.

Firm-level financial and innovation results refer to the overall organizational benefits that are realized when there is an increase in productivity and workflow improvements as well as at the financial level. At the firm level, the organization seeks to determine whether the implementation of AI technologies results in benefits such as improved operational efficiency, lower development costs, faster time-to-market, or new forms of digital products/services, among others. Empirical studies conducted to determine the impact of AI technology adoption by organizations revealed that artificial intelligence technologies could drive organizational growth through innovation and improved productivity (Babina et al., 2024). Therefore, it is critical to understand the role of organizational culture and leadership in the successful implementation of AI coding assistants, given the fact that the achievement of a financial return is dependent on the organization's ability to adopt the new technology.

The majority of the literature is found to have been limited to the first tier of the ROI model. However, the second and third tiers of the model have been found to be underdeveloped in the context of AI coding assistants. This underdevelopment of the second and third tiers of the model points to the fact that the effective usage and integration of AI coding assistants have been found to be mediating factors in the ROI model.

2.3 Measuring ROI in AI Contexts: Financial, Operational, and Strategic Dimensions

Return on investment (ROI) is not absolute in nature but rather relative. ROI is generally described as net benefits earned divided by total costs incurred over a particular period. For AI initiatives, costs generally include subscription fees, integration costs, training costs, governance costs, different models used to mitigate risks, and workflow costs.

The potential benefits that could be earned from implementing AI are as follows:

2.3.1 Time saved translates to reduced labour costs.

A major advantage of AI-based coding assistants is the potential for these tools to decrease the time it takes for a programmer to do their job. If the time it takes to do development tasks is decreased, it implies that the labour costs associated with the development process will decrease. According to experimental research conducted on generative AI, the availability of AI-based assistance has a significant impact on the time it takes to complete a given knowledge-intensive task without compromising the quality of the output (Noy & Zhang, 2023). Similar results have been obtained by other researchers in the field of software engineering, where it was found that the availability of AI-based assistance for the development process can accelerate the development process (Peng et al., 2023).

2.3.2 Increased productivity.

The other possible benefit associated with the use of AI coding assistants is related to the potential for enhancing development throughput. Throughput refers to the volume of work that a development team can process within a particular period. The potential for enhancing development throughput means that an organization can process a larger volume of software features, bug fixes, or updates within a single development cycle without a corresponding increase in the number of developers. Studies on the use of AI-based work tools indicate that generative AI tools have the potential to increase output levels substantially within a particular period, a factor that allows workers to process more deliverables within a particular period (Noy & Zhang, 2023).

2.3.3 Defects reduced.

The coding assistants that AI offers may be beneficial in improving the quality of software code. Defect rates refer to the number of defects or errors that are associated with a given software. It is suggested that when a developer is offered coding assistance and intelligent coding suggestions, he or she may be able to avoid certain coding defects. Research studies have shown that AI-assisted software development is beneficial to developers since it offers coding accuracy and assistance in solving coding challenges (Peng et al., 2023).

2.3.4 Reduction in rework costs.

Rework occurs when software development teams are required to dedicate additional resources to either correcting errors, re-coding some of the features sometimes might be the whole project, and correcting defects that are detected after the development process or version upgrades or when the team has to write the software in a modern language. It is one of the key factors that contribute to costs in software development, considering its effect on time and its ability to cause delays in software development. By improving coding precision and offering real-time support to software developers, there is an opportunity for coding assistants to reduce the need for rework. The timely detection of errors in the development process allows software development teams to avoid the significant costs that are likely to result from addressing errors in subsequent software development phases.

2.3.5 Faster time to market.

Another way in which the use of AI coding assistants may help reduce the time it takes to deliver new software products or features is through the reduction of the time to market. Time to market refers to the length of time it takes to develop a given software product or feature from the start of the project to the release of the feature to the market. The faster the time to market, the better the organization can respond to market opportunities. The empirical literature has proven that the use of AI in the business environment leads to faster innovation processes in the business environment (Babina et al., 2024).

2.3.6 Increased revenues via innovation.

Finally, the adoption of AI has the potential to lead to organizational growth through innovation. This is because when an organization can build new features more easily and faster or try new

ideas more easily, it has the potential to create more revenue opportunities through digital products or services. The adoption of AI has the potential to lead to innovation-driven growth. The study on the adoption of AI by firms revealed that firms that adopted AI had a positive relationship between product innovation and financial performance (Babina et al., 2024). This means that an AI coding assistant has the potential to lead to not only increased operational efficiency but also increased competitiveness of an organization.

Empirical research on AI adoption in organizations suggests that AI capability is a factor in innovation-driven growth (Babina et al., 2024). For quantifying the potential benefits that could be earned from the adoption of AI in team-level software development, a more refined measurement of AI benefits is required.

In the software development domain, cycle time, deployment frequency, defect density, and mean time to recovery act as a framework to assess the overall performance in the software development domain. These could be easily translated into monetary values using the cost of delay, defect costs, and resource allocation.

The research on AI has generally used perception-based productivity metrics. Even in experiments, researchers have focused on short-term results rather than long-term monetary results (Noy & Zhang, 2023).

2.3.7 The ideal model for calculating ROI on AI should include:

- **Performance Metrics**

The objective engineering performance metrics are the measurable parameters used to evaluate the performance and productivity of software development teams. The importance of the performance metrics is undeniable since it provides empirical evidence of the performance of the AI coding assistants. The performance metrics used include the development cycle time, the deployment frequency, the defect density, and the mean time to recovery. The performance metrics are widely used to evaluate the performance of the software development process since it encompasses the productivity and quality aspects of the process. The performance of the software development process is essential since it provides evidence of the effectiveness of the AI coding assistants. For instance, the use of generative AI tools has been observed to improve the performance of the task completion speed and the quality of the outcome, which may be reflected in the performance of the software development process (Noy & Zhang, 2023). The

importance of the performance metrics is undeniable since it provides empirical evidence of the performance of the AI coding assistants.

- **Cost Calculations**

Full cost accounting is a comprehensive measure of all costs associated with the adoption and maintenance of artificial intelligence technology. In terms of AI coding assistants, organizations must consider all costs beyond the costs associated with the software tool. These costs may include costs associated with training developers, tool integration, updating policies and procedures, and maintaining the environment that is necessary to support AI usage. There are also costs associated with oversight that organizations must consider when using AI technology. Research studies have shown that organizations tend to underestimate all costs associated with technology adoption when studies are based solely on direct technology costs (Babina et al., 2024). To properly assess ROI, all costs associated with AI adoption must be considered. These costs include direct and indirect costs.

- **Quality of organizational adoption**

Organizational adoption quality relates to the extent and quality of the incorporation of AI tools in organizational routines and practices. It should be noted that the mere incorporation of AI coding assistants in the development environment does not guarantee the quality and extent of the usage of such tools in organizational settings. Organizational adoption quality of AI tools can be regarded as high if developers effectively incorporate AI tools in coding activities, testing activities, and collaborative activities such as code reviews. Empirical studies of the organizational adoption of AI tools suggest that the organizational benefits of AI technologies depend significantly on the extent and quality of the incorporation of these tools in organizational routines and activities and the extent of organizational learning (An et al., 2024). Therefore, organizational adoption quality can be regarded as a critical factor in determining the organizational value of AI coding assistants.

- **Governance Costs**

Governance costs refer to the costs associated with managing and regulating the usage of artificial intelligence (AI) technologies in an organization. The usage of AI coding assistants poses various governance challenges to an organization in terms of data privacy, intellectual property rights, software quality assurance, and compliance with organizational policies. According to

various scholarly research on the governance of AI, effective governance mechanisms are necessary to ensure that the usage of AI coding assistants is within acceptable risk parameters and complies with organizational standards (Schneider et al., 2023). However, implementing governance mechanisms requires time and administrative costs, which are additional costs to consider when evaluating the return on investment in artificial intelligence.

- **Risks Involved**

The risk exposure adjustments relate to how potential risks associated with the adoption of AI are considered when evaluating financial returns. There is a potential risk of incorporating coding assistants that may lead to software defects, security breaches, or copyright issues. The adoption of AI is likely to increase software defects, security breaches, or copyright issues. This implies that an organization is likely to incur additional costs when risks associated with AI adoption are realized. According to an empirical study on organizational adoption of AI, technological advantages need to be balanced with risk management and governance to ensure sustainable value creation (Schneider et al., 2023). The necessity of adapting the return on investments formula to account for risks involved in the adoption of artificial intelligence technology means that one should be cautious and avoid likely exaggeration of potential benefits.

In the field of research on AI-coded assistants, there is no unified model. This paper aims to minimize the gap in knowledge.

2.4 Organizational Culture as a Mediator of AI Value Realization.

Thus, organizational culture plays a significant role in the interpretation, trust, and institutionalization of AI tools. Digital organizational culture has been proven to enhance the link between AI capabilities and innovation performance (An et al., 2024). In other words, the moderating effect shows that organizational culture plays a significant role in the payoff of technological capabilities.

Trust formation is a significant factor for the adoption of AI tools. Daly et al. (2025) proved that the formation of trust with AI tools is a function of organizational experience and the presence of support systems and governance systems. Excessive trust is a threat to the adoption of AI tools, while a lack of trust is a barrier to the adoption of AI tools.

Psychological safety is a significant factor for the adoption of AI tools in the context of software engineering teams. In other words, the presence or absence of psychological safety is a significant factor for the transformation of AI suggestions into quality code.

While organizational culture is a significant factor for the adoption of AI tools, it is not quantified in the context of the AI ROI model. Most studies on the impact of organizational culture on AI adoption qualitatively address the impact of organizational culture on AI adoption. Thus, the present study proposes a quantified approach to organizational culture as a mediating factor for the impact of AI coding assistants on the ROI.

2.5 Leadership and AI Governance.

The leadership function assumes critical importance in ensuring that AI adoption is in sync with business goals and risk management processes. The body of research on AI governance has identified three interrelated but distinct layers of AI governance (1) data governance, (2) model governance, and (3) organizational/system governance (Schneider et al., 2023). However, leadership ensures that AI governance structures are embedded within the organization rather than being adopted in an informal and ad hoc manner.

The leadership function assumes importance in defining return on investment (ROI) in AI in the following areas:

2.5.1 Executive sponsorship.

Leadership has an important role to play in ensuring the achievement of value from artificial intelligence technology. Research studies on the digital transformation have all proven the importance of leadership in the achievement of the objectives of technology adoption. Among the most important leadership constructs in the achievement of technology adoption objectives is executive sponsorship, which refers to the support role played by the organizational leadership in the achievement of the objectives of technology adoption. When the organizational leadership provides support to the objectives of technology adoption, it becomes easier to integrate emerging technology, such as the use of artificial intelligence coding assistants, into the organizational strategy instead of it being used as a standalone productivity tool by individual developers (Schneider et al., 2023). Further research studies on artificial intelligence technology have proven the importance of the organizational leadership in the achievement of organizational

preparedness to adopt digital innovations and the ability of employees to integrate artificial intelligence technology into their work practices (An et al., 2024).

2.5.2 Resource allocation.

Another important role of a leader involves resource allocation, which includes the provisioning of the financial, technical, and human resources needed to support the effective integration of the various AI tools into the organizational processes. The effective adoption of AI tools in the organizational processes is more than just the acquisition of the various AI tools. For instance, empirical research on the economic impact of the adoption of AI tools has shown that the ability of the organization to invest in the acquisition of various capabilities would have a positive impact on the ability of the organization to realize productivity improvements from the various AI tools (Babina et al., 2024). In the case of software engineering, the allocation of resources may limit the ability of the developer to experiment with the various coding assistants, thus limiting the ability to realize the various learning processes.

2.5.3 Metrics alignment.

The role of leadership in achieving the alignment of metrics, which means the congruence of the organizations performance metrics and expected results of employing AI tools, cannot be overestimated. AI technologies, including coding assistants influence many aspects of software development productivity such as speed of software development, quality of code produced, collaboration, and defect reduction. The research in productivity results from generative AI technologies indicates that, while these tools are helpful in speeding up the accomplishment of tasks, the outcomes depend heavily on the assessment of the output and its integration in the organizational activities (Noy & Zhang, 2023). In situations where speed of results is set as a goal and sufficient quality is not achieved, the application of these technologies sometimes leads to an increase in the amount of technical debt. However, metrics that balance speed with quality outcomes allow an organization to leverage the performance outcomes of generative AI coding assistants to achieve operational performance.

2.5.4 Policy enforcement.

Another key responsibility of leaders in this case is policy enforcement, which is described as the development and implementation of governance systems that guide the usage of AI coding

assistants in the workflow of an organization. The coding assistants pose new challenges in terms of governance, including intellectual property rights and verification of the source of the code, and managing risks to security. The research on AI governance shows that when an organization has a system of governance in place, it is in a better position to manage technological risks and sustain performance improvements with the adoption of AI coding assistants (Schneider et al., 2023). This reduces uncertainty among developers and provides guidance on the usage of AI coding assistants.

2.5.5 Ethical considerations.

Finally, the ethical considerations associated with the use of artificial intelligence need to be addressed by the leaders. In the case of generative artificial intelligence, ethical considerations arise regarding transparency, ownership of intellectual properties, data privacy, and bias. Ethical governance practices ensure the effective implementation of artificial intelligence technology in a manner that is consistent with the regulatory framework and ethical codes of practice. Ethical artificial intelligence practices help to build trust relationships with employees, which is a critical factor in the long-term institutionalization of artificial intelligence technology. Empirical research studies on the organizational responses to artificial intelligence technology have established the importance of ethical governance mechanisms in building trust relationships between employees and artificial intelligence technology, which is essential to sustain the integration of artificial intelligence technology into the organization (Daly et al., 2025)

If leadership were to be absent in AI adoption, then AI coding assistants would be used in an optional capacity rather than being used to drive business performance. In addition to this, it would also help in reducing risks in intellectual property and code authenticity.

However, research studies that have investigated leadership in AI adoption and its direct linkage with financial ROI in AI coding assistants are scarce. The body of research has primarily been cantered on AI governance principles but has seldom been able to quantify financial ROI.

This has led to the need to incorporate leadership as a key mediating variable in AI ROI modeling.

2.6 South African Context and Emerging Market Considerations.

The adoption of artificial intelligence (AI) occurs in an institutional context characterized by defined differences in capabilities and limited availability of resources in emerging markets. In South Africa, for example, members of the South African software engineering sector agree on the fact that the competition and changes in labor conditions resulting from AI adoption are favorable (Poisat et al., 2024).

The software engineering sector in South Africa includes many big companies that use complex systems and need digital transformation. Nevertheless, the governing authority and specialization of the workforce differ in many organizations. This highlights the importance of leadership and organizational culture in the achievement of AI benefits. In addition, on resource-scarce environments, justifying the return on investment (ROI) of AI implementation becomes essential. In this case, misalignment in the organization increases the risk of implementation.

Despite the surge in interest in the adoption of AI in South African software engineering teams, there still exists a scarcity of peer-reviewed literature on the ROI of using AI coding assistants. This highlights the importance of this paper.

2.7 Synthesis and Research Gap.

Nonetheless, modern research shows that generative AI models increase task-level productivity (Noy & Zhang, 2023). Additionally, investments in AI have been shown to be associated with business growth and innovation results (Babina et al., 2024). However, responses in the market to AI adoption are not uniform due to the presence of risks of AI adoption, including AI integration (Lui et al., 2021). Also, the effectiveness of AI models depends on the company's culture in the digital environment (An et al., 2024). The importance of AI governance is defined since it necessitates the necessary compliance of leadership and strategies (Schneider et al., 2023). In addition to that, the context of the South African case is subject to limitations (Poisat et al., 2024).

Nevertheless, some critical knowledge gaps remain:

- **Insufficient return on investment modelling for AI-based coding assistants**

The current literature on the subject suggests that AI coding assistants can improve the productivity of developers. However, there is a lack of literature on the financial return on investment of AI coding assistants. Most of the literature focuses on the productivity of developers at the task level or the perceived usefulness of AI tools. For example, experimental studies have found that generative AI tools significantly improve the efficiency of task performance and the quality of outputs in a knowledge work setting (Noy & Zhang, 2023). Other studies on AI programming tools have found that developers can perform coding tasks quickly with the aid of AI tools (Peng et al., 2023). However, there is a lack of literature on the financial return on investment of AI coding assistants.

- **Inadequate inclusion of culture and leadership within the financial model**

Organizational culture and leadership have been recognized as key factors in the achievement of successful technology implementation outcomes. The empirical findings have suggested that the organizational culture for digitalisation can strengthen the impact of the capabilities of artificial intelligence on innovation and performance (An et al., 2024). Leadership has been observed as the key factor in the development of the governance structure for the implementation of responsible AI in the organizational environment (Schneider et al., 2023). However, the importance of organizational culture and leadership has been recognized in the literature while studying these factors in isolation for the achievement of technological performance outcomes. Only limited empirical research has been conducted on the development of the organizational culture and leadership factors within the financial framework for the assessment of the return on investment in AI technologies.

- **Insufficient measurement of the operation at team level**

The available literature on AI-driven productivity tends to focus more on individual performance measures and not team-level measures to ensure operational effectiveness. However, it is to be noted that software development is a team-based activity in which developers interact with each other through various processes. Studies that have been conducted to understand the role of AI coding assistants have shown that these tools are used at various stages of the software development lifecycle, though there is a significant variation in their usage (Sergeyuk et al., 2025). It is not clear how these tools influence the productivity and quality of software engineering teams without analyzing team-level measures.

- **Lack of empirical research from the continent**

Most of the research on AI adoption and its organizational outcomes is based on empirical research undertaken in North America, Europe, and parts of Asia. Therefore, there is a scarcity of empirical research on how these AI technologies impact organizational outcomes in Africa. Research on the impact of artificial intelligence on the workforce in South Africa has seen a surge of interest in artificial intelligence adoption while underscoring the need to conduct more research on its impact on organizational outcomes within the local environment (Poisat et al., 2024). The lack of literature on AI programming assistants in teams is a hindrance to our perception of the role of contextual factors in determining the returns on investment of AI usage.

2.8 Chapter summary

This chapter reviewed evidence on the productivity and organizational value of generative AI and AI coding assistants. The literature indicates that while AI coding tools can boost productivity at an operational level, it does not imply that these productivity improvements will result in a positive return on investment for organizations. Achieving the desired value realization will depend on the method of artificial intelligence implementation in the workflow and the accompanying organizational capabilities.

The review found that leadership, organizational culture, governance, and the quality of implementation are critical factors in securing the conversion of productivity enhancements into organizational benefits. Critically, limitations of current ROI metrics were identified, given that they usually do not pay enough attention to outcomes related to workflow and organization level.

Four critical gaps were uncovered, namely insufficient research into financial ROI of AI coding assistants, poor application of leadership and organizational culture in ROI models, lack of team level evidence, and absence of empirical studies from South Africa. This study seek to address these gaps by examining how IT leadership and organizational culture relate to AI coding assistant assimilation and ROI within South African software engineering teams.

Chapter 3

Research Methodology

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Chapter 4

Results

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Chapter 5

Discussion

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Chapter 6

Conclusion

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Appendix A

Ethical Clearance Certificate

**CBE RESEARCH ETHICS COMMITTEE**

Dear Researchers

ETHICAL APPROVAL GRANTED FOR RESEARCH PROJECT

Decision: Clearance granted

This letter serves to confirm that the proposed research project indicated in the table below, has been reviewed by the Department of Applied Information Systems at the University of Johannesburg. Ethical clearance is hereby granted and is valid for three years, from 01 January 1900 until 31 December 2999.

Applicant	
Supervisor	
Student/staff number	
Title	
Decision date at meeting	
Reviewers	
Ethical clearance code	
Rating of application	

CODE 01 – Approved

CODE 02 – Approved with suggestions/requirements
with no re-submission

CODE 03 – Referred back

CODE 04 – Disapproved, cannot re-submit

The researcher/s may now commence with the study providing that:

1. The researcher/s will ensure that the project adheres to ethical research requirements
2. The researcher/s will be conducting the study as set out in the approval application
3. The researcher/s will ensure that project adheres to all applicable legislation, scopes of practice, professional codes of conduct and scientific standards as it pertains to the field or study.
4. The researcher/s will bring under the attention of the research ethics committee any proposed changes, concerns that arise, and unexpected ethical management issues
5. Any changes that can affect the study-related risks for participants or researchers must be reported to the committee in writing
6. Only data collected during the three valid period is permissible and valid for the study

7. No data collect before or after the three valid period is ethically valid and the Research Ethics Committee takes no responsibility for any issues arising in this situation
8. No fieldwork activities may continue after the ethical clearance has expired. A request for an extension of ethical clearance can be made in writing to the REC
9. All data collection must be done within the Government regulations of the lockdown
10. The researcher/s must be aware that COVID-19 may influence the responses and needs to be controlled for

Furthermore, the researcher/s must please address the following comments from the ethics committee:

- N/A

Prof M Name

AIS REC

aisrec@uj.ac.za

Appendix B

Appendix Chapter

B.1 Appendix section

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Appendix C

Another Appendix Chapter

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